

Beyond Point-Attached Semantics: Stable Object-Centric Semantic Fields for Robust Manipulation

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Abstract—Point-cloud policies for robotic manipulation must operate on RGB-D observations whose samples vary with viewpoint, occlusion, depth noise, segmentation, and sensor configuration. Although semantic enrichment can reveal functional object parts, features attached directly to observed points remain coupled to these observation-specific samples, resulting in unstable policy inputs. We propose a stable object-centric semantic field that uses an observed object point cloud as support and reads part-aware embeddings at resampled 3D query locations. While the query locations are drawn from the observed object region, their semantics are inferred from the support-conditioned continuous field rather than attached to individual support samples. The field is trained from category-level part annotations using part anchoring, cross-instance alignment, and support-perturbation consistency, and is then frozen to export semantic point clouds as additional conditioning for a point-cloud policy. Experiments on four RoboTwin simulation tasks and four real-world bimanual manipulation tasks show that our representation provides more consistent functional-part features and improves policy success over raw point clouds, 2D feature lifting, and discrete 3D point-wise features, with larger margins over these baselines in the real-world evaluation. These findings highlight the value of organizing policy inputs around functional object structure, suggesting that object-conditioned semantic fields can help policies learn manipulation behaviors that transfer across object instances despite variability in their observations.

Index Terms—Object-Centric Representation Learning; Continuous Semantic Fields; Robust Robotic Manipulation

I. INTRODUCTION

ROBOTIC manipulation is fundamentally grounded in three-dimensional geometry. A policy must determine where an object can be grasped, which region should make contact, and how the end effector or a tool should move relative to the object. These decisions often depend on functional parts rather than object identity alone: a mug handle supports grasping and hanging, a hammer handle and head serve different roles, and the opening and spout of a container constrain stirring and pouring. Because the geometry of these parts varies across instances, successful manipulation requires a representation that preserves spatial structure while exposing functional correspondence across objects.

Point clouds provide a natural 3D interface for this purpose and have enabled effective visuomotor policies [1], [2]. In

practice, however, a point cloud is not the object surface, but a partial measurement produced by a particular camera arrangement, viewpoint, depth sensor, calibration and sampling procedure. Even when the physical scene is unchanged, the observed point set can vary because of self-occlusion, missing depth, reflective materials, mask errors, and random resampling. These effects become more pronounced in real-world deployment than in simulation. A policy trained directly on such observations must therefore learn the manipulation mapping while simultaneously resolving functional-part ambiguity and suppressing observation-specific variation, often from a limited number of demonstrations.

A natural response is to enrich observed geometry with semantic features, by lifting 2D foundation-model descriptors into 3D [3]–[5] or using point-wise 3D semantic encoders [6], [7]. Such features expose functional regions, but semantic enrichment does not automatically remove observation dependence. Coordinates and their attached descriptors can still change together, requiring the policy to resolve correspondences across both object instances and measurements of the same instance.

This limitation motivates separating observation from semantic readout. We use the observed object cloud as a *support set* conditioning a continuous field and evaluate embeddings at independently resampled *query locations*. Queries remain observation-dependent rather than fixed canonical points. The decoupling lies in decoding each embedding from the complete support-conditioned field instead of attaching it directly to a support sample. Training under perturbed observations encourages corresponding object regions to retain similar semantic responses.

We instantiate this idea as a category-level semantic field learned with part supervision. A frozen 3D encoder provides support features for an object-conditioned tri-plane cache, and a coordinate-conditioned decoder produces an embedding at each query. Part supervision identifies functional regions, cross-instance alignment associates corresponding parts, and consistency regularization encourages reduced sensitivity to changes in the support observation. After training, the frozen field exports query coordinates and embeddings as a semantic point cloud for a DP3-style policy. This interface preserves explicit 3D structure and adds part-aware information without changing the policy objective or action representation.

We evaluate the method from both representation and manipulation perspectives. Representation analysis examines whether the learned field produces consistent part-aware features across object observations and instances, while ablation experiments assess the contribution of its main components

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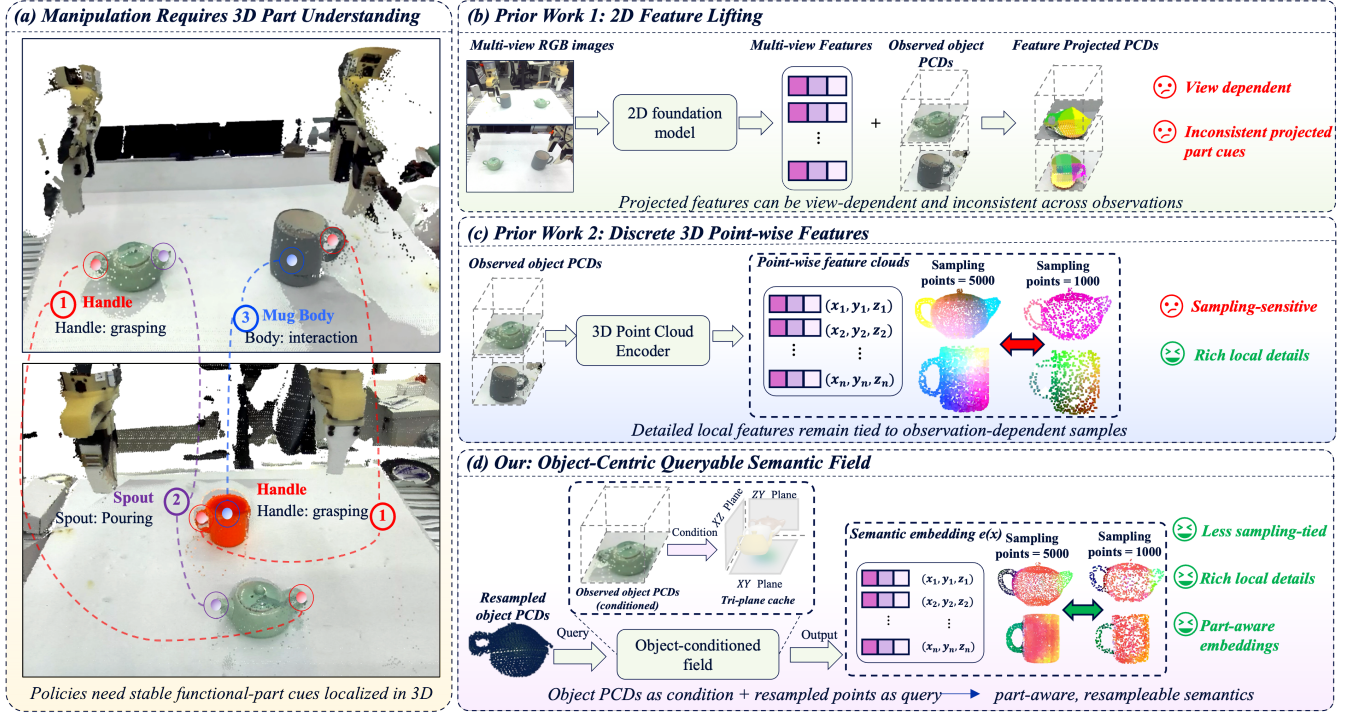


Fig. 1: Teaser of our object-centric continuous semantic field. The point-attached 2D and 3D baselines attach semantics to observation-dependent samples. We instead condition a queryable semantic field on the object point cloud and read part-aware embeddings at explicit 3D query locations, producing semantic point clouds for downstream policy learning.

and training objectives. We measure downstream policy performance on four RoboTwin simulation tasks [8] and four real-world bimanual manipulation tasks. Together, these experiments test the quality of the learned representation and its practical value for policy learning. Our contributions are threefold:

- We formulate a support/query-decoupled object representation that reads functional-part features through an object-conditioned continuous field instead of attaching them directly to noisy support samples. The formulation preserves a point-cloud interface while allowing semantic readout at resampled 3D locations.
- We learn category-level fields using part anchoring, cross-instance alignment, and support-perturbation consistency, and qualitatively examine part discrimination and cross-instance consistency. Simulation ablations assess the three learning objectives and the query sampling design.
- We integrate the resulting semantic point clouds into a DP3-based policy with an unchanged action objective and evaluate manipulation in simulation and on a real bimanual platform, including held-out real-world object instances.

II. RELATED WORK

A. 3D Visuomotor Policy Learning

Imitation learning acquires manipulation skills from visual and proprioceptive demonstrations through behavioral cloning [9], [10] or diffusion-based action generation [11]. Recent policies make spatial structure explicit using point clouds [1], [2], voxel grids [12], or geometrically related

multi-view features [13]. Point clouds are particularly attractive because they preserve metric coordinates through a compact policy interface. However, raw point clouds do not identify functional parts, and RGB-D samples vary with viewpoint, occlusion, segmentation, and resampling. Existing policies typically absorb this variability through augmentation or stronger encoders. Our work instead keeps the point-cloud policy and action objective fixed while improving the object representation supplied to it.

B. Semantic 3D Representations for Manipulation

Semantic 3D representations supplement geometry with functional cues. One family of methods lifts pretrained 2D descriptors such as DINOv2 [14] into 3D, including Distilled Feature Fields [15], G3Flow [3], and GenDP [5]. Another family learns affordances, action trajectories, or dense part-aware features directly from 3D data [6], [7], [16], [17]. Large-scale point-cloud encoders [18], [19] and fine-grained part datasets such as PartNext [20] further support transferable geometric and part understanding. These approaches show that semantic features can expose task-relevant structure that geometry alone may not reveal. Nevertheless, descriptors evaluated only on currently observed points remain coupled to their visibility and sampling. We instead use part supervision to learn an object-conditioned function from which semantic embeddings are queried for policy input.

C. Queryable Object-Centric Fields

Object-centric correspondences provide a structured basis for category-level manipulation. kPAM represents

manipulation-relevant keypoint affordances [21], TAX-Pose estimates task-specific cross-object transformations [22], and Neural Descriptor Fields provide continuous SE(3)-equivariant descriptors [23]. Dense feature fields, including D3Fields [24], Distilled Feature Fields [15], GenDP [5], and G3Flow [3], further demonstrate the utility of queryable 3D semantics for learned manipulation. Our method shares the continuous-query formulation but targets the coupling between semantic descriptors and noisy observed samples. The observed object points condition the field, while the policy input is produced at resampled query locations. Category-level part objectives organize these queried embeddings around functional regions. We analyze the resulting representation and evaluate it in simulated and real-world manipulation without changing the downstream policy formulation.

III. METHODOLOGY

Figure 2 outlines two stages: learning a support-conditioned semantic field from part annotations and using the frozen field to represent objects for manipulation.

A. Problem Formulation

At time t , a manipulation policy observes a scene point cloud $\mathcal{C}_t$, robot proprioception q_t , and segmented point clouds $\{\mathcal{P}_t^k\}_{k=1}^K$ for K task-relevant objects, and predicts an action chunk A_t :

$$A_t \sim \pi(\cdot \mid \mathcal{C}_t, q_t, \{R(\mathcal{P}_t^k)\}_{k=1}^K), \quad (1)$$

where R converts each object observation into a policy-facing representation. An object cloud is a partial measurement: changes in viewpoint, depth quality, segmentation, and sampling can alter its coordinates and attached features even when the object state is unchanged. We do not explicitly estimate these sensing factors. Instead, we seek an object representation that preserves 3D locations and functional-part information while reducing sensitivity to observation variation.

We realize R with a support-conditioned semantic field. The observed object points form a support set $\mathcal{P}_{\text{sup}}$, while an independently resampled set $\mathcal{Q} = \{x_j\}$ determines where policy-facing semantics are read:

$$f_\theta(x \mid \mathcal{P}_{\text{sup}}) = (e_x, \ell_x), \quad R(\mathcal{P}_{\text{sup}}) = \{(x, e_x) : x \in \mathcal{Q}\}, \quad (2)$$

Here, f_θ denotes the complete field, with θ collecting all trainable parameters; $e_x \in \mathbb{R}^d$ is an L2-normalized embedding and $\ell_x \in \mathbb{R}^C$ contains training-time part logits. Queries need not be fixed across observations: each embedding is decoded from the support-conditioned field rather than attached to one support sample. We train a separate field per category on part-annotated PartNext models [20], then freeze it for policy learning. The policy receives query coordinates and embeddings, but not part logits or labels.

B. Support-Conditioned Semantic Field

Given an object observation $\mathcal{P}_{\text{sup}}$, the field first constructs an object-conditioned cache and then decodes semantics at arbitrary queries. During field training, we sample $N_s = 5,000$

support points uniformly with respect to mesh surface area and $N_q = 2,048$ labeled surface queries using 75% part-balanced and 25% area-weighted sampling. At deployment, both sets instead come from the partial observed object cloud. In either case, we subtract the support centroid from both sets, divide their coordinates by the maximum support radius, and translate them so that the center of the support's xy bounding box and its minimum z coordinate lie at the origin.

We voxelize the normalized support at a grid size of 0.01 and provide its coordinates, colors, and normals to a frozen Utonia encoder E [19]. We use its final output restored to the input support resolution, without decoder-level feature upcasting. A lightweight learnable adapter g_ϕ maps each point feature into the field feature space:

$$\tilde{\mathcal{F}}_{\text{sup}} = g_\phi(E(\mathcal{P}_{\text{sup}})). \quad (3)$$

The adapter consists of two linear layers with 256 channels, each followed by LayerNorm and GELU. It projects frozen encoder descriptors to the cache width and lets the field objectives adapt them to part semantics without updating Utonia. Its parameters ϕ belong to θ . We then construct:

$$\mathcal{T}_{\text{obj}} = \Pi(\mathcal{P}_{\text{sup}}, \tilde{\mathcal{F}}_{\text{sup}}) = \{T_{xy}, T_{xz}, T_{yz}, F_{\text{glob}}\}, \quad (4)$$

where T_{xy} , T_{xz} , and T_{yz} are 64×64 feature maps with 256 channels, and $F_{\text{glob}} \in \mathbb{R}^{256}$ is the mean adapted support feature. We normalize each axis to the support bounding box with 5% padding, then bilinearly splat support features onto the three coordinate planes.

For a query x , we use the same padded support bounds to project its coordinate onto each plane and bilinearly interpolate the corresponding features. The three 256-D samples are concatenated and fused by a linear-LayerNorm-GELU block into $F_{\text{loc}}(x) \in \mathbb{R}^{256}$. A two-layer 256-D MLP receives the query coordinate, local feature, and replicated global feature, and predicts

$$(e_x, \ell_x) = h_\psi(x, F_{\text{loc}}(x), F_{\text{glob}}). \quad (5)$$

Here, h_ψ is the query decoder within f_θ , with $\psi \subset \theta$; the full field comprises support encoding, cache construction, and query decoding. The decoder produces a 128-D L2-normalized embedding and applies a linear classifier for part logits. Local features preserve spatial detail, while the global feature supplies instance context.

C. Part-Aware Field Learning

The field must satisfy two complementary requirements: its embeddings should identify functional parts across object instances, and they should remain stable when the same object is observed differently. We therefore train it using paired augmented observations and three complementary losses.

Paired observation construction. We transform each normalized support-query pair into two augmented observations:

$$(\mathcal{P}_{\text{sup}}^{(v)}, \mathcal{Q}^{(v)}) = \mathcal{T}_{\xi_v}(\mathcal{P}_{\text{sup}}, \mathcal{Q}), \quad v \in \{1, 2\}, \quad (6)$$

where $\mathcal{T}_{\xi_v}$ independently applies a uniform SO(3) rotation and Gaussian coordinate jitter with standard deviation 0.005 to both sets. Each view is an augmented observation, not an

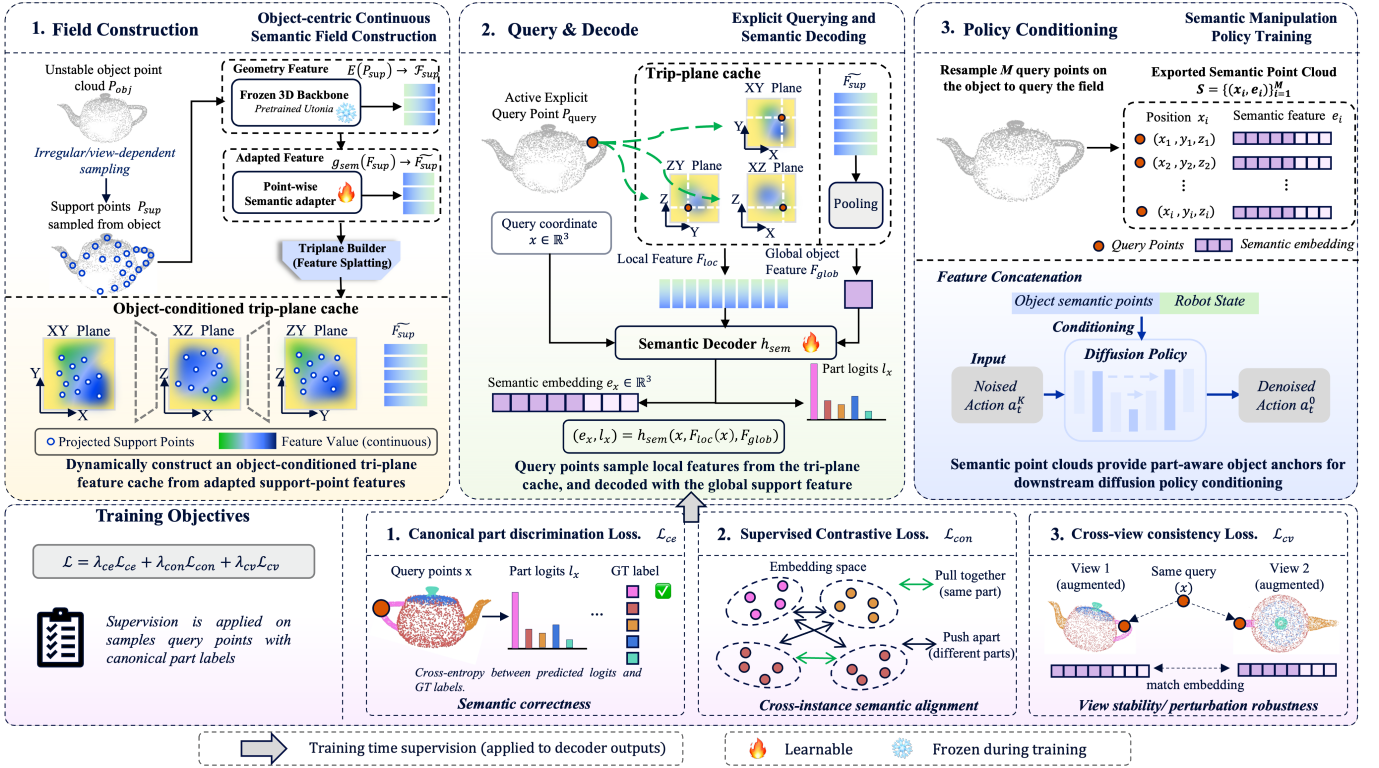


Fig. 2: Overview of our object-centric continuous semantic field. Support points condition an object-specific tri-plane cache; 3D query locations read out part-aware embeddings and training-time part logits. The frozen field is then queried at resampled object locations to export semantic point clouds for downstream manipulation policies.

additional camera. Shared surface samples and labels establish query correspondences despite transformed coordinates, allowing us to encourage semantic invariance. These augmentations model pose and local coordinate changes, but not occlusion, missing depth, or segmentation errors.

Part anchoring. We apply part-label cross-entropy to the query logits in both views:

$$\mathcal{L}_{ce} = \frac{1}{2N_q} \sum_{v=1}^2 \sum_{j=1}^{N_q} \text{CE}(\ell_j^{(v)}, y_j). \quad (7)$$

This loss supervises functional-part prediction at each query. To also encourage corresponding parts to occupy nearby locations in the embedding space used by the policy, we introduce a complementary alignment objective.

Cross-instance semantic alignment. We therefore apply supervised contrastive learning independently in each view and average the results. For an embedding e_i , let $P(i)$ contain other queries in the minibatch with the same part label, including queries from different object instances. With temperature τ ,

$$\mathcal{L}_{con} = -\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(e_i^\top e_p / \tau)}{\sum_{a \neq i} \exp(e_i^\top e_a / \tau)}, \quad (8)$$

where $\mathcal{I}$ contains anchors with at least one positive. We use $\tau = 0.1$ and retain at most 2,048 valid queries per view. This loss pulls embeddings of the same functional part together and separates different parts, directly structuring the representation consumed by the policy.

Paired-augmentation consistency. The contrastive loss aligns part-level clusters but does not explicitly require the

same surface location to produce the same response under different observations. Using the known query correspondences, we minimize

$$\mathcal{L}_{cr} = \frac{1}{N_q} \sum_{j=1}^{N_q} \left(1 - \langle e_j^{(1)}, e_j^{(2)} \rangle\right). \quad (9)$$

Because the embeddings are L2-normalized, this cosine loss directly penalizes semantic changes at corresponding locations. Unlike $\mathcal{L}_{con}$, which establishes category-level part structure, $\mathcal{L}_{cr}$ specifically reduces sensitivity to the modeled observation transform.

Overall objective. We jointly optimize

$$\mathcal{L} = \lambda_{ce}\mathcal{L}_{ce} + \lambda_{con}\mathcal{L}_{con} + \lambda_{cr}\mathcal{L}_{cr}, \quad (10)$$

with $(\lambda_{ce}, \lambda_{con}, \lambda_{cr}) = (1.0, 0.2, 0.1)$. We train only the semantic adapter, local fusion block, and decoder for 4,000 epochs with minibatches of six object instances, using AdamW with learning rate 10^{-3} and weight decay 10^{-4} . Utonia remains frozen, and training uses mixed precision.

D. Semantic Point Clouds for Manipulation

After field training, all representation modules are frozen. At each observation frame, we independently sample 1,024 support points and $M = 256$ query locations from the observed object cloud $\mathcal{P}_t^k$, using replacement only when fewer valid points are available. After joint normalization and field evaluation, we restore the query coordinates to the world frame and instantiate the representation in Eq. (1) as

$$R(\mathcal{P}_t^k) = \mathcal{S}_t^k = \{(x_i, f_{\theta}^{\text{emb}}(x_i | \mathcal{P}_{\text{sup},t}^k))\}_{i=1}^M. \quad (11)$$

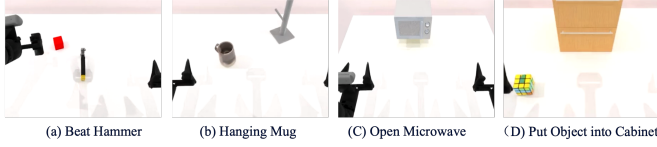


Fig. 3: The four RoboTwin tasks: Beat Hammer, Hanging Mug, Open Microwave, and Put Object into Cabinet.

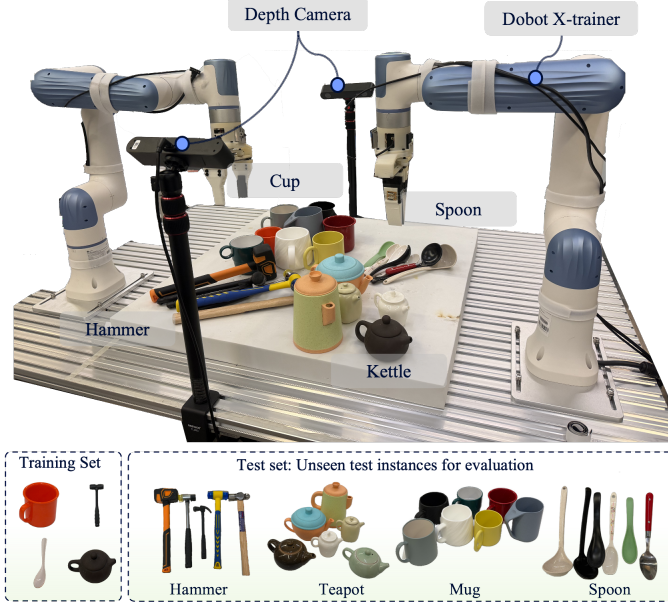


Fig. 4: Real-world setup on the DOBOT X-Trainer dual-arm platform. The bottom row distinguishes demonstration objects (Training Set) from held-out evaluation objects (Test Set), which differ in geometry and appearance.

The field is evaluated once per incoming observation. We use DP3 [1] as the policy backbone. A 1,024-point scene cloud and each 131-channel semantic cloud (XYZ plus the 128-D embedding) are processed by separate max-pooled PointNet encoders, each producing a 128-D feature. These features are concatenated with proprioception and supplied to the unchanged diffusion objective and action representation.

IV. EXPERIMENTS

We conduct a series of experiments to evaluate the proposed representation from both representation and manipulation perspectives. We aim to answer the following questions:

- (1) **Representation Robustness:** Do feature visualizations exhibit part discrimination and consistent patterns across object instances in simulated and real observations?
- (2) **Manipulation Effectiveness:** Does the resulting representation improve policy performance across the evaluated simulation tasks and held-out real-world object instances?
- (3) **Component Contributions:** How do support/query separation and the three field-learning objectives affect downstream manipulation success?

A. Experimental Setup

a) Simulation setup.: We evaluate four RoboTwin tasks [8] spanning single-arm and dual-arm manipulation:

Hanging Mug hangs a mug on a rack through its handle; *Beat Hammer* grasps the handle and strikes a target with the hammer head; *Open Microwave* pulls the door handle; and *Put Object into Cabinet* grasps the cabinet handle and opens the door. These tasks require localization of functional parts rather than object identity alone (Fig. 3). We follow the standard RoboTwin data split without modifying the benchmark data partitioning or processing protocol. All methods use the same demonstrations and test instances. Each policy is trained for 3,000 epochs on 50 demonstrations and evaluated for 100 episodes per seed.

b) Real-world setup.: We further evaluate four bimanual tasks on a DOBOT X-Trainer dual-arm platform equipped with two ZED 2i RGB-D cameras, as shown in Fig. 4. For each task, we collect 50 teleoperated demonstrations using designated training objects and evaluate on held-out instances with different geometry and appearance. The tasks are (1) *Grasp Mug*, grasping the mug handle and placing the mug stably on the target base; (2) *Beat Cube*, grasping the hammer handle and striking the cube with the hammer head; (3) *Stir Mug*, grasping the mug handle with the left hand and the spoon handle with the right hand, then stirring inside the mug with the spoon head; and (4) *Pour Water*, grasping the kettle handle and pouring water from its spout into the target mug. These functional-part requirements define task success. Each method is evaluated for 20 trials per task under matched initial conditions, and we report the success rate.

c) Baselines.: **DP3** [1] is our primary geometry-only baseline. **DINOV2 Lifting** is a controlled baseline inspired by G3Flow [3] and D³Fields [24]: frozen DINOV2 [14] descriptors are sampled at image projections of observed object points and lifted into 3D. **Utonia Point-wise** attaches frozen Utonia [19] descriptors directly to observed points, without field construction or resampled queries. We additionally compare with **G3Flow** [3] in simulation, using its complete semantic-flow pipeline rather than treating DINOV2 Lifting as its reproduction.

d) Implementation and evaluation details.: Grounded SAM [25] segments task-relevant objects. Masked depth pixels are back-projected from calibrated views, transformed into the robot/world frame, and fused into object clouds. DINOV2 Lifting, Utonia Point-wise, and ours share these masks and reconstruction steps, as well as a DP3-based interface: separate PointNets encode the scene and semantic object clouds, and their features are concatenated with proprioception. The policy objective and action representation are unchanged. Point budgets follow Sec. III-D.

For G3Flow, we follow the official implementation and deployment procedure, including object-model-based semantic-flow construction and FoundationPose tracking to update feature locations as objects move. Unlike per-observation DINOV2 Lifting, this pipeline maintains object-aligned features through pose tracking. We retain these method-specific components rather than forcing G3Flow into the controlled lifting interface.

All comparisons use matched task demonstrations and evaluation episodes. DP3, DINOV2 Lifting, Utonia Point-wise, and ours additionally share observation histories and training

TABLE I
COMPARATIVE EVALUATION ON FOUR ROBOTWIN TASKS (SUCCESS RATES IN %).

Method	<i>Hang Mug</i>	<i>Beat Hammer</i>	<i>Open Micro.</i>	<i>Put Cabinet</i>	Average
DP3 [1]	23.0 $\pm$ 4.0	72.0 $\pm$ 2.0	62.0 $\pm$ 3.0	72.0 $\pm$ 2.0	57.3
DINOV2 Lifting	23.0 $\pm$ 7.0	78.0 $\pm$ 5.0	58.0 $\pm$ 3.0	68.0 $\pm$ 4.0	56.8
Utonia Point-wise	31.0 $\pm$ 4.0	77.0 $\pm$ 2.0	65.0 $\pm$ 1.0	76.0 $\pm$ 1.0	62.3
G3Flow [3]	25.0 $\pm$ 5.0	78.0 $\pm$ 5.0	64.0 $\pm$ 2.0	78.0 $\pm$ 3.0	61.3
Ours	37.0 $\pm$ 3.0	84.0 $\pm$ 1.0	73.0 $\pm$ 1.0	83.0 $\pm$ 1.0	69.3

schedules. Policies are trained for 3,000 epochs with batch size 128 on a single NVIDIA RTX 4090 GPU, also used for inference. Simulation reports mean success and standard deviation over training seeds, with 100 episodes per seed; real-world results report successes out of 20 trials per task.

B. Simulation Experiments

Table I reports the controlled representation comparisons and the complete G3Flow baseline. Our method achieves the highest success rate on every task, averaging 69.3%: 12.0 percentage points above DP3, 7.0 above Utonia Point-wise, and 8.0 above G3Flow. These gains across handle localization, tool-part discrimination, and articulated interaction support the utility of the learned semantic cues.

DINOV2 Lifting improves Beat Hammer but matches or underperforms DP3 on the other three tasks, yielding an average of 56.8%. Its larger standard deviations on several tasks indicate greater variability across training runs; these statistics alone do not identify multi-view fusion as the cause. Utonia Point-wise improves the DP3 average by 5.0 percentage points, while our learned field adds a further 7.0 points. These comparisons support the utility of the learned semantic representation, but do not separately identify the effects of field construction and additional part supervision. The ablations below examine the learning objectives and query sampling design.

a) *Comparison with G3Flow.*: G3Flow averages 61.3%, improving on DINOV2 Lifting by 4.5 percentage points. It improves three tasks and matches Beat Hammer, but remains below ours on all four. Its standard deviations (2–5 points) are higher than ours (1–3 points) on every task, although no higher than DINOV2 Lifting. During evaluation, we observed drift in FoundationPose estimates, which can displace object-aligned semantic features and destabilize policy inputs. This is a plausible contributor to the remaining variability, not an isolated causal finding. Our field instead reads semantics from each current support observation without propagating tracked object poses. The comparison therefore evaluates complete pipelines rather than isolating the effect of tracking alone.

C. Real-World Experiments

Figure 5 shows representative task executions on held-out objects. Table II reports success under the matched evaluation protocol. Our method achieves the highest success rate on every task and an average of 67.5%, compared with 35.0%

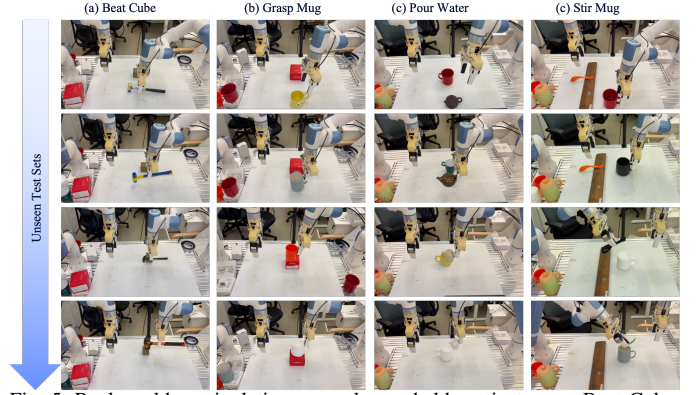


Fig. 5: Real-world manipulation examples on held-out instances: Beat Cube, Grasp Mug, Pour Water, and Stir Mug.

TABLE II
COMPARATIVE EVALUATION ON FOUR REAL-WORLD TASKS (SUCCESS RATES IN %). Each method is evaluated for 20 trials per task.

Method	<i>Grasp Mug</i>	<i>Beat Cube</i>	<i>Stir Mug</i>	<i>Pour Water</i>	Average
DP3 [1]	35.0	40.0	15.0	20.0	27.5
DINOV2 Lifting	35.0	45.0	25.0	25.0	32.5
Utonia Point-wise	35.0	45.0	35.0	25.0	35.0
Ours	85.0	85.0	50.0	50.0	67.5

for Utonia Point-wise, 32.5% for DINOV2 Lifting, and 27.5% for DP3. This corresponds to gains of 32.5 and 40.0 percentage points over the strongest semantic baseline and the geometry-only baseline, respectively. The advantage is most pronounced on *Grasp Mug* and *Beat Cube*, where our method succeeds in 17 of 20 trials, compared with 7 and 9 trials for Utonia Point-wise.

The margin over Utonia Point-wise is larger in the real world than in simulation (32.5 versus 7.0 percentage points). Depth noise, partial visibility, calibration residuals, and segmentation errors may contribute by destabilizing point-attached features. Our field is designed to reduce such variation through object-conditioned readout. However, the evaluations use different tasks and object distributions, so the gap cannot be attributed solely to sensing noise. Section IV-D examines relevant qualitative feature patterns.

D. Representation Analysis

We qualitatively examine whether the representations preserve both *intra-instance part discrimination* and *cross-instance part consistency*. Feature embeddings are projected to RGB with PCA for visualization. For each method and object category, a single PCA basis is fitted jointly to all displayed instances and then applied without instance-specific normalization. Colors are therefore comparable across instances within the same method-category group: consistent colors at corresponding functional parts indicate aligned embeddings, whereas color changes across instances reveal representation drift. Absolute colors should not be compared across different methods or categories because their PCA bases are fitted independently.

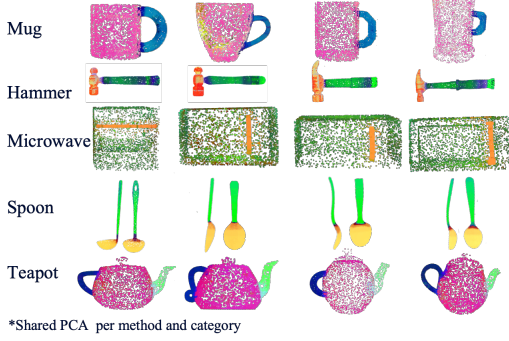


Fig. 6: Part-aware field embeddings queried on simulation instances from five object categories. Within each category, all instances share one PCA projection. Corresponding functional parts exhibit consistent colors across instances, while distinct parts remain clearly separable.

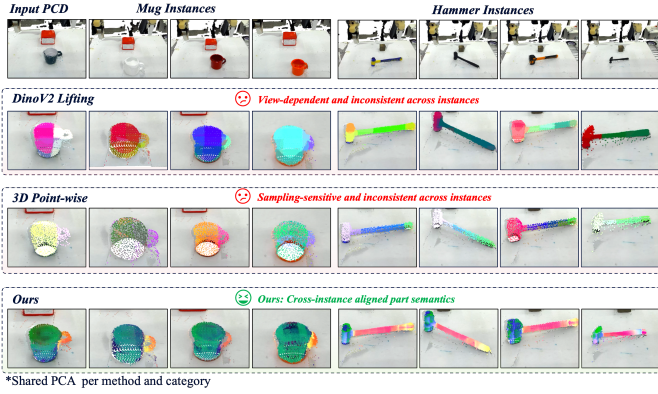


Fig. 7: Cross-instance feature comparison on real observations of mugs and hammers. DINOv2 lifting is view-dependent, and point-wise 3D descriptors remain sensitive to observed sampling. Our field preserves aligned part-level patterns across geometrically and visually different instances.

a) *Simulation representation.*: Figure 6 visualizes embeddings across five categories with substantial shape variation. Corresponding regions retain similar colors: mug handles remain distinct from bodies, and hammer heads from handles. The other categories exhibit analogous part distinctions. These patterns suggest cross-instance consistency while preserving functional detail within each object.

b) *Real-world representation.*: Figure 7 compares DINOv2 Lifting, Utonia Point-wise, and our field under physical RGB-D observations. DINOv2 produces meaningful local structure within an individual observation: it can often separate a mug handle from its body or a hammer head from its handle. However, the same functional part changes color markedly across instances and viewpoints, and some regions exhibit abrupt, spatially nonuniform responses. Thus, its descriptors are locally discriminative but not reliably aligned across objects of the same category after multi-view lifting. Utonia Point-wise reduces the view dependence of the 2D features and preserves recognizable geometry, yet corresponding parts still exhibit color variation across the displayed instances. These visualizations do not isolate whether that variation is caused by shape, sampling, or other observation differences.

Our field retains coherent handle/body and head/handle patterns across the four mug and four hammer instances despite geometry and appearance differences. This combines part

TABLE III
ABLATION STUDY OF REPRESENTATION DESIGN AND FIELD-LEARNING OBJECTIVES. Average success rate (%) across four simulation tasks.

Variant	S/Q Sep.	$\mathcal{L}_{ce}$	$\mathcal{L}_{con}$	$\mathcal{L}_{cr}$	Average (%)
Coupled support/query		✓	✓	✓	66.7
Without $\mathcal{L}_{ce}$	✓		✓	✓	61.3
Without $\mathcal{L}_{con}$	✓	✓		✓	64.7
Without $\mathcal{L}_{cr}$	✓	✓	✓		68.9
Full model	✓	✓	✓	✓	69.3

discrimination with cross-instance consistency, in agreement with the policy gains in Table II. Controlled perturbation tests are still needed to establish the contribution of sensing noise.

E. Ablation Study

We isolate the contribution of the support/query separation and the three objectives used for semantic-field learning. All variants retain the frozen Utonia encoder, DP3 policy backbone, demonstrations, point budget, and optimization settings. The coupled support/query variant preserves the complete field and training objective, but selects its 256 policy-facing query coordinates directly from the sampled support set instead of independently resampling them from the observed object cloud. The remaining variants retain the full support/query-decoupled architecture and each remove exactly one loss from Eq. (10). We report the average success rate across the four simulation tasks.

Table III reports 69.3% for the full model and 66.7% for coupled support/query, a 2.6-point difference. Both variants retain field-based decoding and all three objectives, so this comparison tests query sampling, not a field against a point-wise model with matched part supervision.

Removing $\mathcal{L}_{ce}$ reduces success to 61.3%, the largest drop (8.0 points), supporting explicit functional-part anchoring. Removing $\mathcal{L}_{con}$ yields 64.7%, a 4.6-point drop, supporting alignment of corresponding parts across instances. Without $\mathcal{L}_{cr}$, success is 68.9%; this 0.4-point difference does not establish a reliable benefit without variability estimates. These ablations measure downstream success, not cross-view feature consistency. Controlled perturbation tests are needed to assess robustness directly. Part anchoring and alignment thus provide the largest observed gains, complemented by independent query sampling.

V. DISCUSSION AND LIMITATIONS

Our method relies on category-level PartNext annotations and a consistent part taxonomy, and currently learns a separate semantic field for each object family. It therefore does not provide open-vocabulary or category-free transfer, while its annotation and training costs grow with the number of object categories. Discrete canonical part labels are also less suitable for deformable objects and ambiguous, continuous, or task-dependent functional regions. At deployment, the field conditions on segmented partial object clouds and does not require observing the complete object surface. Nevertheless, its predictions may degrade under severe occlusion, complete invisibility of a functional part, sparse or corrupted depth, and segmentation failure. Finally, the representation captures part-aware geometric and semantic structure but does not explicitly model physical properties, object dynamics, or task-conditioned affordances. Future directions include shared fields, weaker supervision, and uncertainty estimation.

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VI. CONCLUSION

We presented an object-centric continuous semantic field that separates the points used to observe an object from the locations where policy-facing semantics are read. Conditioned on a partial object point cloud, the field produces part-aware embeddings at resampled query locations and exports them as a semantic point cloud to a DP3-based policy without changing its action objective. Part anchoring and cross-instance alignment organize the field around functional correspondences, while consistency regularization encourages reduced sensitivity to changes in the support observation. Across four RoboTwin tasks and four real-world bimanual tasks, the resulting representation improves manipulation success over raw point clouds, lifted 2D features, and point-wise 3D features, with larger margins over these baselines in the real-world evaluation. Ablations further show that part anchoring and cross-instance alignment provide the dominant training signals, while support/query separation contributes an additional architectural benefit. These results support object-conditioned semantic readout as a reliable interface for robust manipulation under observation and instance variation.

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